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4. Rank and the Rank-Nullity Theorem

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Problem 4.1 Let **A** be the **3** × **5** matrix

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ -1 & -2 & 9 & 10 & 11 \\ 1 & 2 & 9 & 11 & 13 \end{pmatrix}.$$

- 1. Find a basis for **CS** (**A**).
- 2. What are **dim NS** (**A**) and **dim CS** (**A**)?

Solution:

- 1. First we find a row echelon form. Add the first row to the second, and add **−1** times the first row to the third:

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 0 & 0 & 12 & 14 & 16 \\ 0 & 0 & 6 & 7 & 8 \end{pmatrix}.$$

Add **−1/2** times the second row to the third:

$$\mathbf{B} := \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 0 & 0 & 12 & 14 & 16 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

This is in row echelon form.
The pivots of **B** are in the first and third columns.

Basis for **CS** (**A**): first and third columns of **A**, i.e., $\begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ 9 \\ 9 \end{pmatrix}.$

- 2. $\text{dim NS}(\mathbf{A}) = \# \text{ non-pivot columns of } \mathbf{B} = 3.$
 $\text{dim CS}(\mathbf{A}) = \# \text{ pivot columns of } \mathbf{B} = 2.$

Definition 4.2 The **nullity** of **A** is defined as the number

$$\text{nullity}(\mathbf{A}) = \text{dim NS}(\mathbf{A}).$$

The **rank** of **A** is defined as the number

$$\text{rank}(\mathbf{A}) = \text{dim CS}(\mathbf{A}).$$

Rank concept check I

1/1 point (graded)
What can the rank of a **3** by **5** matrix be? (Choose all that apply.)

☒ 0

☒ 1

☒ 2

☒ 3

☐ 4

☐ 5



Solution:

The rank is equal to the dimension of the column space of the matrix. The rank must be less than or equal to **3** because the column space is a subspace of $\mathbb{R}^3$. All of the ranks **0**, **1**, **2** and **3** can be realized as shown by the following matrices:

Rank	Example
0	$\begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$
1	$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$
2	$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$
3	$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}$

The rank of any matrix **A** is always between **0** and **min (m, n)**. To see why, note that the column space is a subspace of $\mathbb{R}^m$, so the rank is less than or equal to **m**. However **CS (A)** is spanned by **n** columns, hence its dimension is less than or equal to **n** and so is rank of **A**. Putting this together, this means that the rank is less than or equal to the minimum of **m** and **n**. To see that every rank between **0** and **min (m, n)** is possible, we can create matrices with these ranks in the same way as for the specific **3 × 5** example above.

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Theorem 4.3 For any **m × n** matrix **A**,

$\text{nullity}(\mathbf{A}) + \text{rank}(\mathbf{A}) = n.$

This theorem is called the **rank-nullity theorem** .

Proof

Let **B** be a row echelon form of **A**.

$$\begin{aligned} \text{nullity}(\mathbf{A}) + \text{rank}(\mathbf{A}) &= \dim \text{NS}(\mathbf{A}) + \dim \text{CS}(\mathbf{A}) \\ &= (\# \text{ non-pivot columns of } \mathbf{B}) + (\# \text{ pivot columns of } \mathbf{B}) \\ &= \# \text{ columns of } \mathbf{B} \\ &= n. \end{aligned}$$

Rank-nullity concept check I

1/1 point (graded)
Which of the following are true? (Choose all that apply.)

- ☐ There exists a **2 × 3** matrix **A** with **dim NS (A) = 0** and **rank (A) = 0**.
- ☐ There exists a **2 × 3** matrix **A** with **dim NS (A) = 0** and **rank (A) = 1**.
- ☐ There exists a **2 × 3** matrix **A** with **dim NS (A) = 0** and **rank (A) = 2**.

☐ There exists a 2×3 matrix $\mathbf{A}$ with $\dim \text{NS}(\mathbf{A}) = 0$ and $\text{rank}(\mathbf{A}) = 3$.

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☒ A 2×3 matrix $\mathbf{A}$ with $\dim \text{NS}(\mathbf{A}) = 0$ does not exist.



Solution:

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Rank-nullity concept check II

1/1 point (graded)

Which of the following are true? (Choose all that apply.)

- ☐ There exists a 3×2 matrix $\mathbf{A}$ with $\dim \text{NS}(\mathbf{A}) = 0$ and $\text{rank}(\mathbf{A}) = 0$.
- ☐ There exists a 3×2 matrix $\mathbf{A}$ with $\dim \text{NS}(\mathbf{A}) = 0$ and $\text{rank}(\mathbf{A}) = 1$.
- ☒ There exists a 3×2 matrix $\mathbf{A}$ with $\dim \text{NS}(\mathbf{A}) = 0$ and $\text{rank}(\mathbf{A}) = 2$.
- ☐ There exists a 3×2 matrix $\mathbf{A}$ with $\dim \text{NS}(\mathbf{A}) = 0$ and $\text{rank}(\mathbf{A}) = 3$.
- ☐ A 3×2 matrix $\mathbf{A}$ with $\dim \text{NS}(\mathbf{A}) = 0$ does not exist.



Solution:

The matrix has 3 rows but only 2 columns, so the rank cannot be greater than 2 . Since $\dim \text{NS}(\mathbf{A}) + \text{rank}(\mathbf{A}) = n$ and $\dim \text{NS}(\mathbf{A}) = 0$, we get $\text{rank}(\mathbf{A}) = n = 2$, so all columns of $\mathbf{A}$ are pivot columns.

An example of such a matrix is the following:

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

When the rank of a matrix equals the number of its columns we say that this matrix has a **full column rank**. Whenever this happens, the matrix has a 0 —dimensional nullspace.

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Rank-nullity concept check III

1/1 point (graded)

The $m \times n$ matrix $\mathbf{A}$ has $\text{rank}(\mathbf{A}) = 2$ and $\dim \text{NS}(\mathbf{A}) = 4$. Which of the following describes all of the possible choices for m and n ?

- ☐ $m = 2$ and $n = 4$
- ☐ $m = 4$ and $n = 6$
- ☐ $m = 6$ and $n = 4$

☒ $m \geq 2$ and $n = 6$

☐ $m \leq 2$ and $n \geq 4$



Solution:

By the rank nullity theorem, $\dim \text{NS}(\mathbf{A}) + \dim \text{CS}(\mathbf{A}) = n$, so the total number of columns is $n = 6$. The rank of $\mathbf{A}$ is dimension of $\text{CS}(\mathbf{A})$, which is a subspace of $\mathbb{R}^m$. So m is at least 2. The following examples that show that each of the values $m \geq 2$ are possible:

- $\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}$
- $\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$
- $\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$
- (Keep adding rows of zeros to get all m !)

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4. Rank and the Rank-Nullity Theorem

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